

7-day and 28-day strength models (independent)

Each model built from its own 30 results with the same protocol · In scale · exhaustive AICc over 716 hierarchical models · generated 2026-09-28

7-day model · 10 parameters · 30 mixtures

$\ln(f_7) \sim \text{RCF} + \text{SS} + \text{Carbonation (4 levels)} + \text{RCF} \times \text{SS} + \text{SS}^2 + \text{Carbonated} \times \text{RCF} + \text{Carbonated} \times \text{SS}$

$\ln f = 2.7343 + 0.0600 \cdot A + 0.7544 \cdot D + 0.0647 \cdot C_{0.5h} - 0.0942 \cdot C_{1h} + 0.0572 \cdot C_{5h} - 0.1071 \cdot A \cdot D - 0.6844 \cdot D^2 + 0.1964 \cdot K \cdot A - 0.3340 \cdot K \cdot D$

$\ln(f_7) = 1.04494 + 0.00835212 \cdot \text{RCF} + 0.0609018 \cdot \text{SS} - 1.4276 \times 10^{-4} \cdot \text{RCF} \cdot \text{SS} - 4.8669 \times 10^{-4} \cdot \text{SS}^2 + 0.104207 \cdot C[0.5 \text{ h}] - 0.0547479 \cdot C[1 \text{ h}] + 0.0966465 \cdot C[5 \text{ h}] + 0.00981798 \cdot K \cdot \text{RCF} - 0.00890739 \cdot K \cdot \text{SS}$

R² / adj / pred
0.95
0.93 / 0.91

RMSE fitted / left out
1.6
2.2 MPa left out

Residual scatter
15 %
replicate batches 19 %

Lack of fit
0.89
vs replicate error (p)

RCF Hierarchy SS Robust Carbonation (4 levels) Hierarchy RCF × SS Moderate SS² Robust Carbonated × RCF Moderate Carbonated × SS Moderate

All selection criteria agree (AICc, BIC, backward elimination at α = 0.10 and 0.05). The carbonation terms are kept in 92 % of residual-bootstrap re-selections but in 26 % of subsamples of 24 mixtures, because they rest on the 8 uncarbonated mixtures.

28-day model · 5 parameters · 30 mixtures

$\ln(f_{28}) \sim \text{RCF} + \text{A/B} + \text{SS} + \text{RCF}^2$

$\ln f = 3.1763 + 0.0757 \cdot A - 0.1462 \cdot C + 0.2510 \cdot D - 0.3293 \cdot A^2$

$\ln(f_{28}) = 4.26393 + 0.0531733 \cdot \text{RCF} + 0.00669398 \cdot \text{SS} - 4.8733 \cdot \text{A/B} - 8.2315 \times 10^{-4} \cdot \text{RCF}^2$

R² / adj / pred
0.54
0.46 / 0.38

RMSE fitted / left out
5.6
6.4 MPa left out

Residual scatter
28 %
replicate batches 20 %

Lack of fit
0.17
vs replicate error (p)

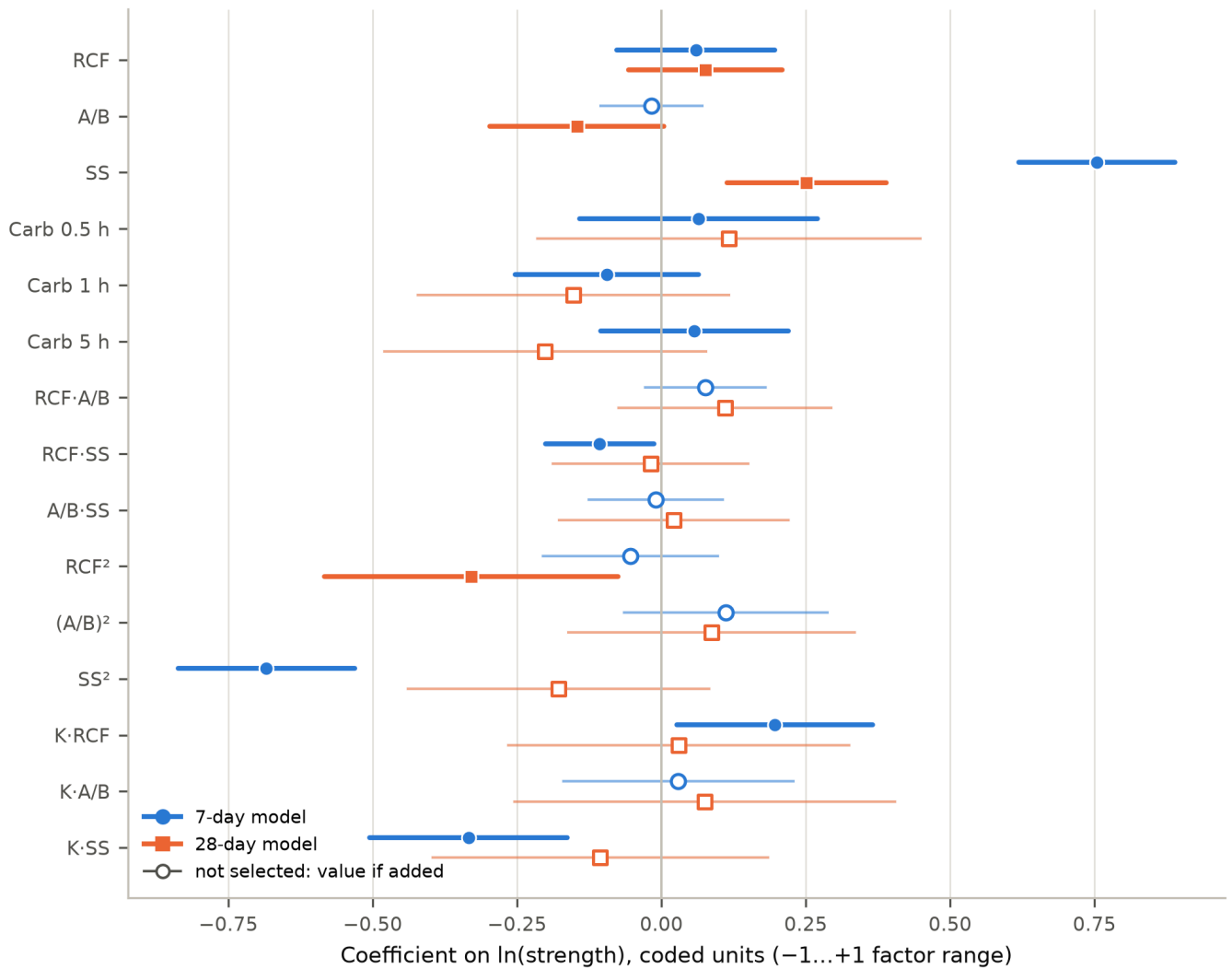
RCF Hierarchy A/B Weak SS Robust RCF² Moderate

SS is the only robust term. A/B depends on mix 7 and RCF² on mix 14: each drops out when that mixture is removed. No carbonation term is detected (best added-term p = 0.17). Predictions for a new mixture are much less precise than at 7 days.

Terms side by side

Candidate term	7-day model: coefficient [95 % CI]	p	Evidence	28-day model: coefficient [95 % CI]	p	Evidence
RCF	0.060 [−0.077, 0.197]	0.37	Hierarchy	0.076 [−0.057, 0.209]	0.25	Hierarchy
A/B	if added: −0.017 [−0.105, 0.071]	0.69	Not detected	−0.146 [−0.297, 0.005]	0.057	Weak
SS	0.754 [0.619, 0.890]	< 0.001	Robust	0.251 [0.113, 0.389]	< 0.001	Robust
Carbonation (4 levels)	Carb 0.5 h: 0.065 [−0.141, 0.271] Carb 1 h: −0.094 [−0.253, 0.065] Carb 5 h: 0.057 [−0.105, 0.220]	0.23	Hierarchy	if added: Carb 0.5 h: 0.117 [−0.215, 0.449] Carb 1 h: −0.152 [−0.422, 0.117] Carb 5 h: −0.201 [−0.480, 0.077]	0.17	Not detected
RCF × A/B	if added: 0.076 [−0.028, 0.180]	0.30	Not detected	if added: 0.110 [−0.074, 0.294]	0.23	Not detected
RCF × SS	−0.107 [−0.201, −0.013]	0.027	Moderate	if added: −0.019 [−0.187, 0.150]	0.82	Not detected
A/B × SS	if added: −0.010 [−0.125, 0.106]	0.91	Not detected	if added: 0.021 [−0.178, 0.220]	0.83	Not detected
RCF ²	if added: −0.054 [−0.205, 0.097]	0.46	Not detected	−0.329 [−0.583, −0.075]	0.013	Moderate
(A/B) ²	if added: 0.112 [−0.064, 0.288]	0.40	Not detected	if added: 0.087 [−0.161, 0.335]	0.48	Not detected
SS ²	−0.684 [−0.837, −0.532]	< 0.001	Robust	if added: −0.178 [−0.439, 0.083]	0.17	Not detected
Carbonated × RCF	0.196 [0.027, 0.366]	0.025	Moderate	if added: 0.030 [−0.265, 0.325]	0.29	Not detected
Carbonated × A/B	if added: 0.029 [−0.170, 0.228]	0.88	Not detected	if added: 0.075 [−0.254, 0.404]	0.27	Not detected
Carbonated × SS	−0.334 [−0.505, −0.163]	< 0.001	Moderate	if added: −0.106 [−0.396, 0.185]	0.24	Not detected

Coefficients of the two independent models, 95 % CI

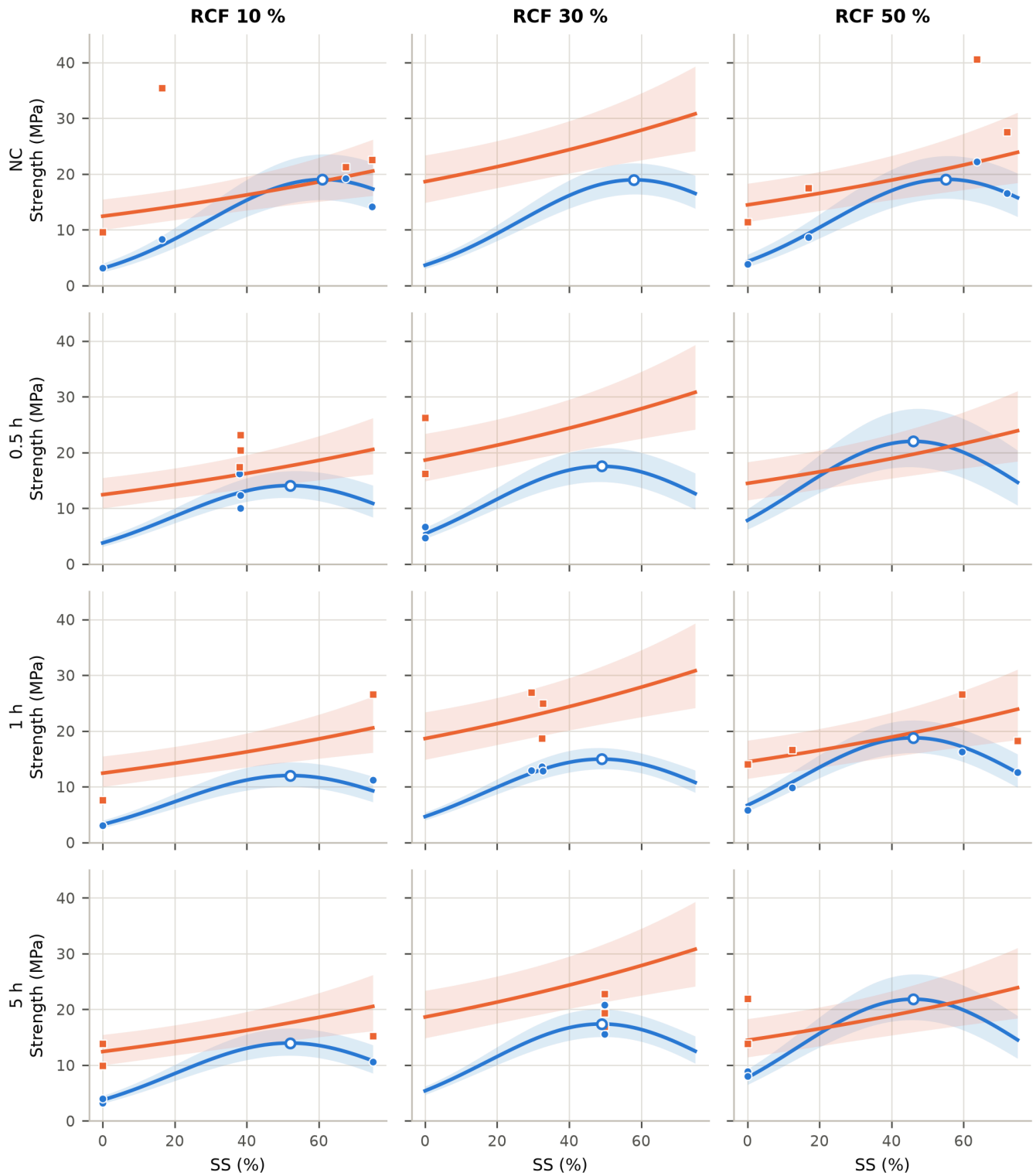


Trends side by side

SS

Carbonation	RCF (%)	7 d: SS at max [95 % boot]	28 d: SS at max	7 d: f(SS 0) / f(max)	28 d: f(SS 0) / f(max)	7 d: f(SS/SH = 2) / f(max)	28 d: f(SS/SH = 2) / f(max)	Suppl.: 28-day data, 7-day terms: SS at max
NC	10	61 % [52-72]	75 % (edge)*	×0.16 [0.12-0.22]	×0.61 [0.46-0.80]	×0.99 [0.95-1.03]	×0.95 [0.92-0.98]	60 %
NC	30	58 % [51-67]	75 % (edge)*	×0.19 [0.15-0.24]	×0.61 [0.46-0.80]	×0.97 [0.91-1.02]	×0.95 [0.92-0.98]	60 %
NC	50	55 % [46-64]	75 % (edge)*	×0.23 [0.17-0.30]	×0.61 [0.46-0.80]	×0.94 [0.87-1.01]	×0.95 [0.92-0.98]	58 %
1 h	10	52 % [42-58]	75 % (edge)*	×0.27 [0.22-0.33]	×0.61 [0.46-0.80]	×0.90 [0.82-0.99]	×0.95 [0.92-0.98]	48 %
1 h	30	49 % [42-54]	75 % (edge)*	×0.31 [0.27-0.36]	×0.61 [0.46-0.80]	×0.86 [0.77-0.96]	×0.95 [0.92-0.98]	47 %
1 h	50	46 % [40-51]	75 % (edge)*	×0.36 [0.29-0.43]	×0.61 [0.46-0.80]	×0.81 [0.71-0.93]	×0.95 [0.92-0.98]	46 %

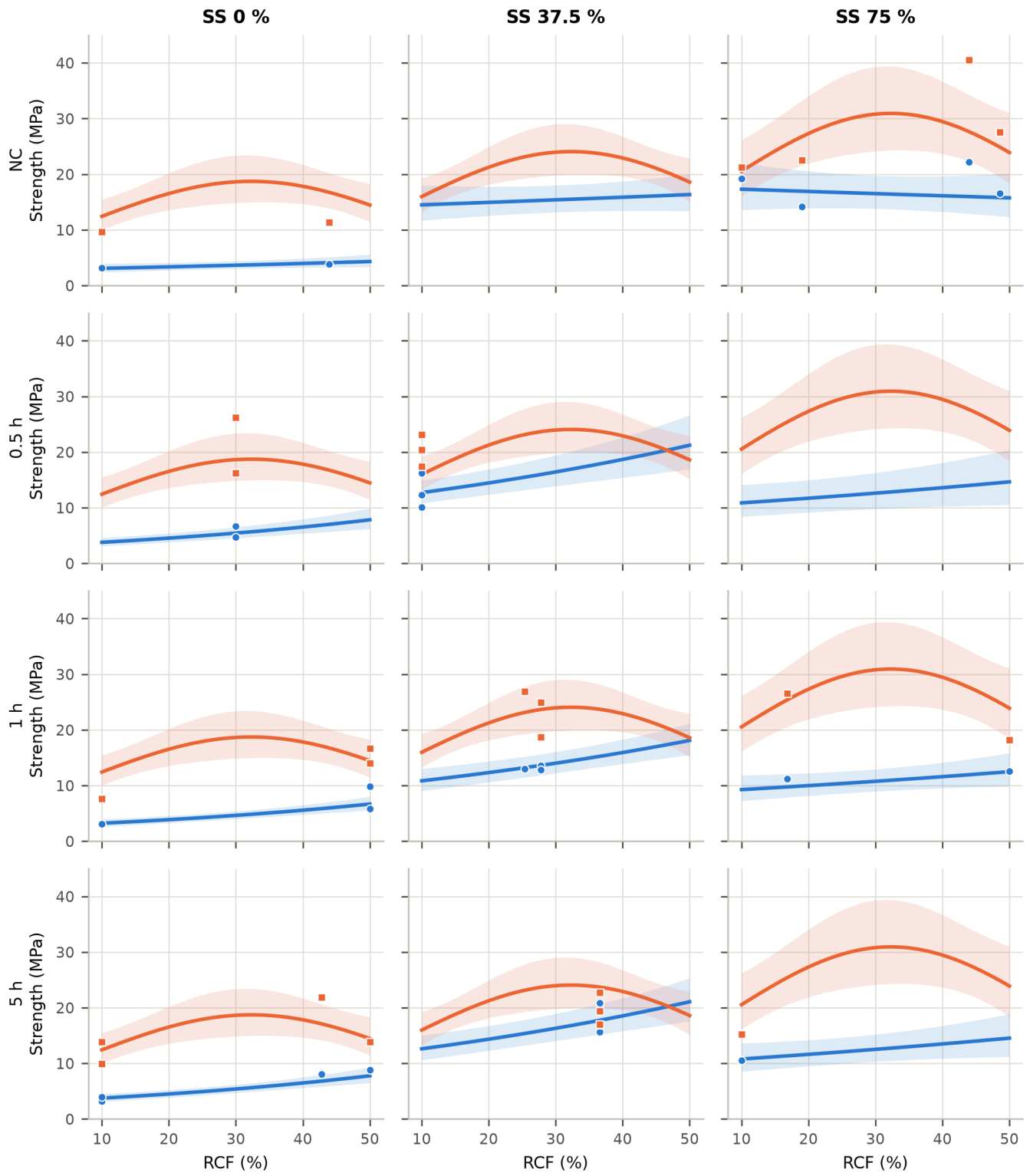
— 7-day model (95 % CI) ● runs within $\pm 10\%$ RCF
 — 28-day model (95 % CI) ○ interior peak



RCF 50 % / 10 %

Carbonation	SS (%)	7-day model	28-day model	Supplementary: 28-day data refitted with the 7-day terms
NC	0	$\times 1.40$ [1.00–1.95], $p = 0.048$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.19$ [0.60–2.33], $p = 0.61$
NC	25	$\times 1.21$ [0.91–1.60], $p = 0.17$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.16$ [0.65–2.06], $p = 0.59$
NC	50	$\times 1.05$ [0.79–1.39], $p = 0.72$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.14$ [0.64–2.02], $p = 0.64$
NC	75	$\times 0.91$ [0.65–1.27], $p = 0.56$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.12$ [0.57–2.20], $p = 0.74$
1 h	0	$\times 2.07$ [1.63–2.62], $p < 0.001$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.39$ [0.86–2.25], $p = 0.17$
1 h	25	$\times 1.79$ [1.48–2.18], $p < 0.001$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.36$ [0.92–2.02], $p = 0.12$
1 h	50	$\times 1.55$ [1.24–1.95], $p < 0.001$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.33$ [0.84–2.11], $p = 0.21$
1 h	75	$\times 1.35$ [0.99–1.83], $p = 0.057$	$\times 1.16$ [0.89–1.52], $p = 0.25$	$\times 1.31$ [0.70–2.45], $p = 0.39$

7-day model (95 % CI) 28-day model (95 % CI) runs within $\pm 15\%$ SS



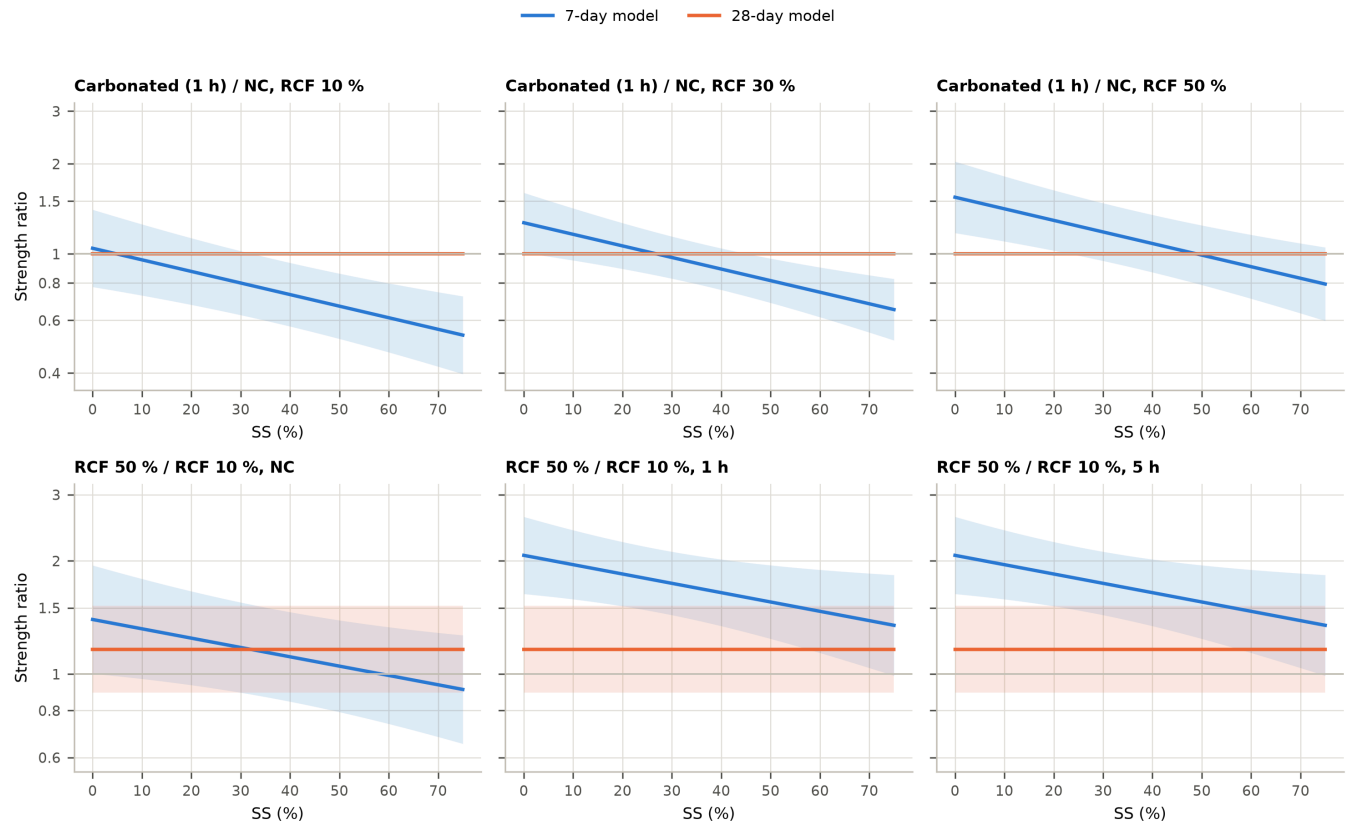
Carbonated (1 h) / NC

RCF (%)	SS (%)	7-day model: 1 h / NC	28-day model: 1 h / NC	Supplementary: 28-day data refitted with the 7-day terms
10	0	$\times 1.04$ [0.78–1.41], $p = 0.76$	1 (not in model)	$\times 0.96$ [0.53–1.77], $p = 0.90$
10	25	$\times 0.84$ [0.65–1.07], $p = 0.15$	1 (not in model)	$\times 0.84$ [0.50–1.40], $p = 0.49$
10	50	$\times 0.67$ [0.52–0.86], $p = 0.003$	1 (not in model)	$\times 0.73$ [0.44–1.22], $p = 0.22$
10	75	$\times 0.54$ [0.40–0.72], $p < 0.001$	1 (not in model)	$\times 0.64$ [0.35–1.18], $p = 0.14$
30	0	$\times 1.27$ [1.01–1.60], $p = 0.042$	1 (not in model)	$\times 1.04$ [0.65–1.67], $p = 0.85$
30	25	$\times 1.02$ [0.86–1.20], $p = 0.83$	1 (not in model)	$\times 0.91$ [0.65–1.28], $p = 0.57$
30	50	$\times 0.81$ [0.69–0.97], $p = 0.020$	1 (not in model)	$\times 0.79$ [0.56–1.12], $p = 0.18$
30	75	$\times 0.65$ [0.51–0.83], $p = 0.001$	1 (not in model)	$\times 0.69$ [0.43–1.12], $p = 0.13$
50	0	$\times 1.55$ [1.18–2.04], $p = 0.003$	1 (not in model)	$\times 1.13$ [0.64–1.98], $p = 0.65$
50	25	$\times 1.24$ [0.99–1.55], $p = 0.063$	1 (not in model)	$\times 0.99$ [0.62–1.56], $p = 0.95$
50	50	$\times 0.99$ [0.79–1.25], $p = 0.93$	1 (not in model)	$\times 0.86$ [0.54–1.37], $p = 0.50$
50	75	$\times 0.79$ [0.60–1.05], $p = 0.10$	1 (not in model)	$\times 0.75$ [0.42–1.33], $p = 0.31$

Carbonation duration

Contrast (RCF 30, SS 37.5)	7-day model	28-day model	Supplementary: 28-day data refitted with the 7-day terms
1 h / 0.5 h	$\times 0.85$ [0.70–1.05], $p = 0.12$	1 (not in model)	$\times 0.81$ [0.54–1.23], $p = 0.31$
5 h / 0.5 h	$\times 0.99$ [0.82–1.21], $p = 0.94$	1 (not in model)	$\times 0.78$ [0.52–1.16], $p = 0.21$
5 h / 1 h	$\times 1.16$ [0.99–1.36], $p = 0.062$	1 (not in model)	$\times 0.96$ [0.69–1.33], $p = 0.79$

A/B 0.42 / 0.48: 7-day model 1 (not in model); 28-day model $\times 1.34$ [0.99–1.81], $p = 0.057$.

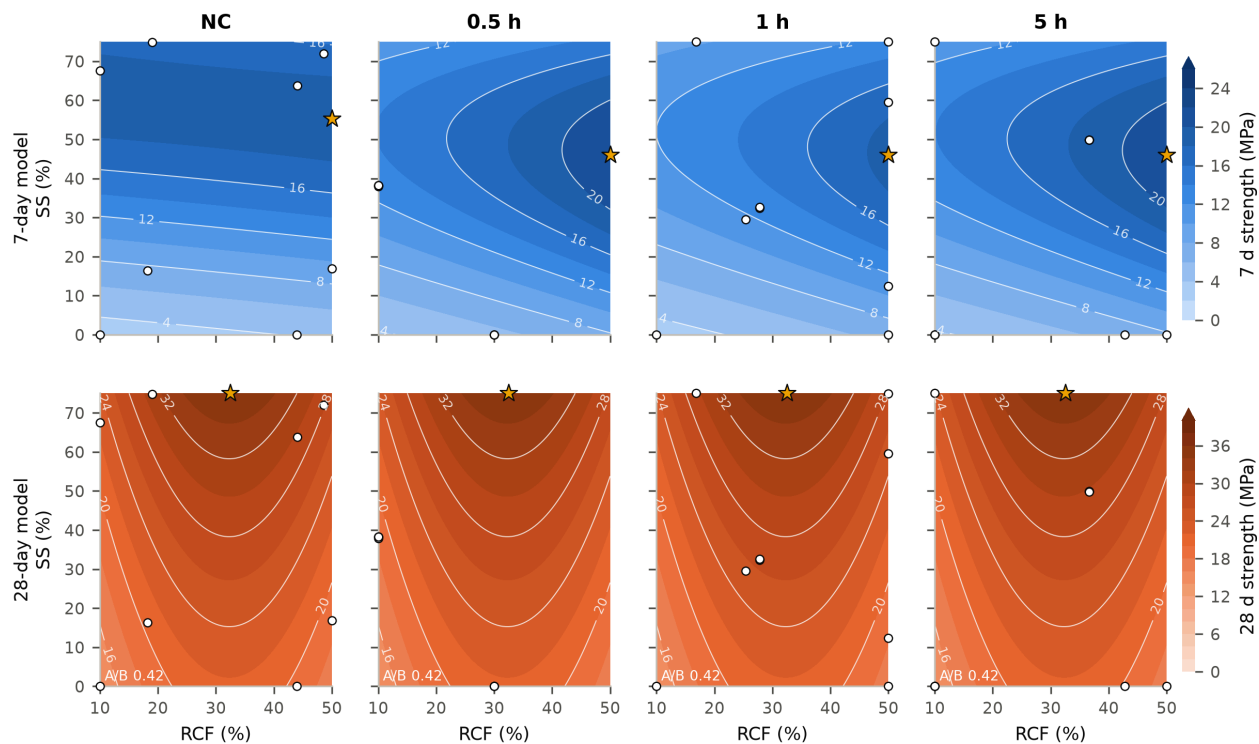


Bands: 95 % CI of the ratio. A ratio of 1 means no effect; a flat line at 1 means the term is not in that model.

Best combination per carbonation level

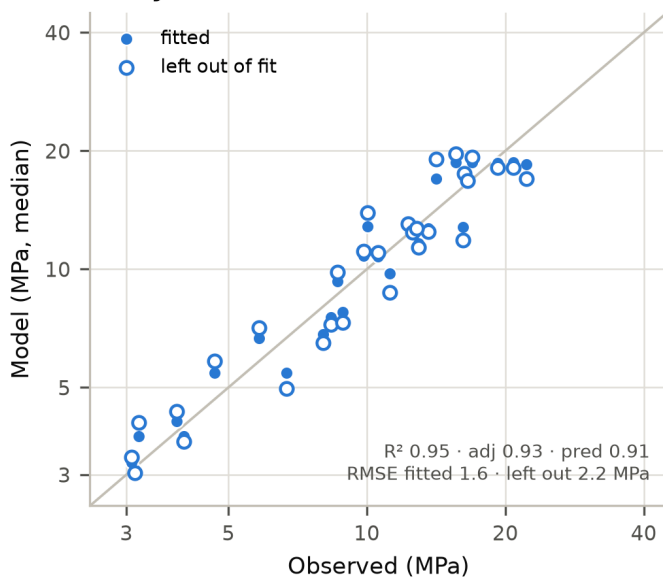
Carbonation	7 d: maximum at	7 d: median [95 % CI] MPa	7 d: 95 % PI	28 d: maximum at	28 d: median [95 % CI] MPa	28 d: 95 % PI
NC	RCF 50, SS 55	19.1 [15.6–23.2]	13.1–27.8	RCF 32, SS 75, A/B 0.42	35.8 [27.2–47.2]	19.0–67.5
0.5 h	RCF 50, SS 46	22.0 [17.4–27.8]	14.8–32.7	RCF 32, SS 75, A/B 0.42	35.8 [27.2–47.2]	19.0–67.5
1 h	RCF 50, SS 46	18.8 [16.1–21.9]	13.2–26.8	RCF 32, SS 75, A/B 0.42	35.8 [27.2–47.2]	19.0–67.5
5 h	RCF 50, SS 46	21.8 [18.1–26.3]	15.1–31.6	RCF 32, SS 75, A/B 0.42	35.8 [27.2–47.2]	19.0–67.5

Median predicted strength over RCF × SS from each model (own colour scale per age). ○ runs at that level, ★ maximum



Validation

a 7-day model



b 28-day model

